Bipartite (k1,k2,l)-sparsity is equivalent to arboricity of the associated butterfly multigraph after adding l-1 edges

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Editorial note. In the source the theorem's own label is written after the statement text (source line 198); the retained span covers the whole theorem environment, label included.

Editorial note. The two citations inside the retained lines (Servatius on k-plane matroids; Nash-Williams on edge-disjoint spanning trees) are resolved by a flattened two-entry bibliography fragment, so no bibliography database and no BibTeX run is needed.

Editorial note. The theorem is stated in the source with k1, k2, l in the positive integers and k1 + k2 > l, although the surrounding discussion also treats l in the nonnegative integers; the statement is retained verbatim and no hypothesis was added.

Editorial note. The retained environments print with this package's own numbering rather than the numbering of the article's full document.

Editorial scope. One original exact result: the equivalence between (k1,k2,l)-sparsity of a bipartite graph and the forest decomposition of its butterfly multigraph after adding any l-1 edges.

Allowed standard prerequisites (editorial).

- Nash-Williams arboricity theorem. A finite multigraph decomposes into k edge-disjoint forests if and only if every nonempty edge subset F' satisfies |F'| <= k(|V(F')| - 1); cited by the author at source line 214.
- Butterfly bookkeeping. The (k1,k2)-fold butterfly graph of a bipartite graph carries k1k2 copies of every edge, so an edge count for the bipartite graph is recovered by summing over the k1k2 pairs of vertex classes.

Definition 1. *Let $G = (V_1, V_2, E)$ be a bipartite graph, with a partition of the vertex set into sets V_1 and V_2, and let $k_1, k_2 \in \mathbb{Z}_{>0}, l \in \mathbb{Z}_{\geq 0}$. Then we say that G is (k_1, k_2, l)-sparse if, for every nonempty subset of $E' \subseteq E$, it holds that*

$$|E'| \leq k_1|V(E') \cap V_1| + k_2|V(E') \cap V_2| - l.$$

We say that G is (k_1, k_2, l)-tight if it is (k_1, k_2, l)-sparse, and if additionally

$$|E| = k_1|V(E) \cap V_1| + k_2|V(E) \cap V_2| - l.$$

To end this section, we will relate (k_1, k_2, l)-sparsity to tree packings. The special case of $(k, 1, k)$-tight and $(1, k, k)$-tight graphs was shown by Servatius [2].

Given a bipartite graph $G = (V_1, V_2, E)$, define its associated (k_1, k_2)-fold butterfly graph B as follows: the vertex set consists of k_1 vertices for each element of V_1 and k_2 vertices for each element of V_2. For each edge $vw \in E$ with $v \in V_1$ and $w \in V_2$, there are edges between all k_1 vertices representing v and all k_2 vertices representing w.

Theorem 2. *Let $k_1, k_2, l \in \mathbb{Z}_{>0}$ with $k_1 + k_2 > l$. A bipartite graph $G = (V_1, V_2, E)$ is (k_1, k_2, l)-sparse if and only if adding any $l - 1$ edges between V_1 and V_2 results in a bipartite graph with an associated (k_1, k_2)-fold butterfly graph $B = (V, F)$ that decomposes into $k_1 k_2$ edge-disjoint forests.*

Proof. Suppose first that any resulting B can be written as the union of $k_1 k_2$ edge-disjoint forests. Then any such B is $(k_1 k_2, k_1 k_2)$-sparse by the Nash-Williams theorem. For any nonempty subsets $V_1' \subseteq V_1$ and $V_2' \subseteq V_2$, adding the $(l-1)$ edges within $V_1' \cup V_2'$, one has, by the sparsity of the resulting graph B, that

$$k_1 k_2 |E(V_1' \cup V_2')| + (l-1)k_1 k_2 \leq k_1 k_2 (k_1 |V_1'|) + k_1 k_2 (k_1 |V_1'|) - k_1 k_2,$$

from which the (k_1, k_2, l)-sparsity of G follows immediately.

Conversely, suppose that we have added any $l - 1$ edges to G, and let B be the resulting butterfly graph. Let $V' \subseteq V(B)$. Note that all edges belonging to $F(V')$ are between $V'_{1,i}$ and $V'_{2,j}$ for some copies of subsets of V_1 and V_2 respectively. Since G is sparse, we can bound $|F(V'_{1,i} \cup V'_{2,j})|$ by sparsity:

$$|F(V'_{1,i} \cup V'_{2,j})| \leq k_1 |V'_{1,i}| + k_2 |V'_{2,j}| - 1$$

Then we have

$$\begin{aligned}
|F(V')| &= \sum_{i=1}^{k_1} \sum_{j=1}^{k_2} |F(V'_{1,i} \cup V'_{2,j})| \\
&\leq \sum_{i=1}^{k_1} \sum_{j=1}^{k_2} \left(k_1 |V'_{1,i}| + k_2 |V'_{2,j}| - 1 \right) \\
&= k_2 k_1 \sum_{i=1}^{k_1} |V'_{1,i}| + k_1 k_2 \sum_{j=1}^{k_2} |V'_{2,j}| - k_1 k_2 \\
&= k_2 k_1 (|V'| - 1).
\end{aligned}$$

The result then follows by the Nash-Williams theorem [1]. □

References

[1] C. St. J. A. Nash-Williams. Edge-disjoint spanning trees of finite graphs. *J. London Math. Soc.* 36 (1961), 445–450.

[2] Brigitte Servatius. *k-plane matroids and Whiteley's flattening conjectures.* In: Combinatorics, graph theory and computing, Springer Proc. Math. Stat. 388, Springer, Cham, 2022, 109–115.